

Paper 2 & Paper 7 Testing Complete + Progress Update

Piter Garcia (RIT Student) <pzg8794@rit.edu>
 Reply-To: pzg8794@rit.edu
 To: Daniel Krutz <dxkvse@rit.edu>

Wed, Jan 7, 2026 at 4:53 PM

Hi Dan,

Thanks for getting back to me—and I appreciate your flexibility as I slowly get back into things.

I'm meeting with Devroop and Sheeraja tomorrow (Thursday) at 2 PM to align next steps. If you're available and would like to join that meeting, I'd be happy to add you. Alternatively, I'm also open to meeting before or after that time, or on Friday if that works better for you.

Also, just to clarify: we do not have two separate Overleaf projects. There's only one shared source—the original project you created ([Overleaf Project](#)). The second link I shared ([Travis version](#)) is an identical copy that I prepared per your suggestion, so that Professor Travis could focus solely on key sections. We haven't received his feedback yet, but now that I'm feeling a bit better, I'll be following up with him today. Would you prefer I give him access to our main version instead, where he can view and comment on every section?

To avoid further back-and-forth, I'll also be updating a Google Doc we've been using to track all communications, feedback, and responses. I'll share that with you shortly to help us stay aligned moving forward.

Let me know what time works best for you!

Best,
 Piter Garcia

On Wed, Jan 7, 2026 at 5:59 AM Daniel Krutz <dxkvse@rit.edu> wrote:
 Hi Piter, glad to hear that you're feeling better.

- I actually don't use that account on slack that you found me on.
- What times would work well for you to zoom this week to circle back with a few things. I am sure that I've forgotten/missed a bunch of your questions over break.
- If I understand things correctly, we currently have 2 overleaf projects with updated content? If so, would it make more sense to have 1 version?

Thanks
 -Dan

On Jan 7, 2026, at 1:29 AM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Hi Dan,

I've been recovering slowly—it's taken longer than expected, but I've finally been able to get out of bed and am now catching up on emails and resuming my work on finalizing my schedule for the semester.

I wanted to clarify something regarding the Overleaf link you sent (<https://www.overleaf.com/project/6945bfa58266e8759ad66f24>)(<https://www.overleaf.com/project/6945bfa58266e8759ad66f24>)). I'm not sure where that link came from, but it doesn't seem connected to any of my projects. As far as I know, there are only two sources:

- * The project you originally created:
<https://www.overleaf.com/project/68ea344896594f27b427ca8f>(<https://www.overleaf.com/project/68ea344896594f27b427ca8f>)
 - * The cleaned-up version I created for Professor Travis (also shared with you):
<https://www.overleaf.com/read/ttmrwxqfjfy#38bfea>(<https://www.overleaf.com/read/ttmrwxqfjfy#38bfea>)
- Both contain the same content.

Also, I've resumed putting together my schedule—especially around GA responsibilities—and wanted to circle back since some of my earlier questions may have been missed during the end-of-semester rush. Let me know if you'd like me to resend anything.

Lastly, I did send you a message on Slack (screenshot below). I was able to find your account in the main RIT workspace under @Daniel Krutz (dxkvse / dxkvse@rit.edu), but not in the MS Data Science channel.

<image.png>

Appreciate your patience, and I'm looking forward to getting fully reconnected. I'll be meeting with Devroop and Sheeraja soon to discuss next steps—let me know if you'd like to be added to those meetings.

Best,
Piter

On Mon, Jan 5, 2026 at 8:50 PM Daniel Krutz <dxkvse@rit.edu> wrote:

Hi Piter, I am just checking in to see how you are doing.

Also, could you give me access (dan7800@yahoo.com) access to <https://www.overleaf.com/project/6945bfa58266e8759ad66f24>

Thanks
-Dan

On Jan 2, 2026, at 9:10 PM, Daniel Krutz <dxkvse@rit.edu> wrote:

I am sorry to hear this. Feel better soon
-Dan

On Jan 2, 2026, at 10:37 AM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Hi Dan,

I will get back to you as soon as I am out of bed, I have been ill in bed for about three days now.

Best,
Piter

On Fri, Jan 2, 2026 at 6:48AM Daniel Krutz <dxkvse@rit.edu> wrote:

Hi Piter, a few things:

1. Could you give me access to this page: <https://www.overleaf.com/project/6945bfa58266e8759ad66f24>
2. Could you send me a message on Slack. My username is [@dxkvse@g.rit.edu](https://rit.slack.com/team/dxkvse)
3. Could you send me a 2-3 sentence overview of how our proposed process works? You don't need to mention results, just give an overview of the process. I'd also add this overview of your process to the abstract of your working version. I think that it just jumps from the proposed problem to your findings? You kind of touch base on your process in I.b of the introduction, but I would double or triple the length of this description as it is pretty high level.

We will touch base next week. How does your Monday morning work, say around 10?

Thanks
-Dan

On Dec 31, 2025, at 3:28AM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Hi Dan,

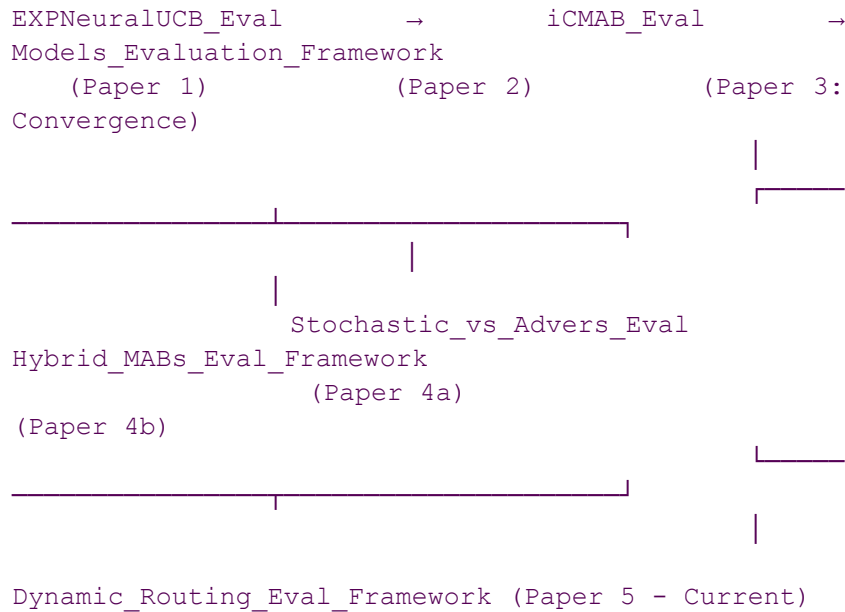
Here are the responses to your questions, organized by how we built the evaluation framework:

Overleaf Document Link

The current working document is this [Overleaf link](#).
There's only one version—I created the second [condensed view](#) (abstract, research questions, results only) per your advice for Professor Travis's feedback.

Framework Evolution & Research Contributions

Our work evolved through a systematic framework-building process:



1. Baseline Replication (Paper 1 – EXPNeuralUCB Framework)

We replicated the quantum routing paper's setup, extending beyond their hardcoded 10K capacity to scaled regimes (SC = 1T, 1.5T, 2T), exposing allocation overfitting in the original design.

2. Predictive Enhancement (Paper 2 – iCMAB Framework)

We incorporated context-aware Pursuit variants (CPursuit, iCPursuit) with topology awareness, neural feature extraction, and world modeling beyond simple reward history.

3. Multi-Model Comparative Framework (Paper 3 – Convergence Point)

Benchmarked 4 algorithms $\times$ 5 attack scenarios $\times$ 4 allocators = 60 conditions:

- **CPursuit**
dominance: 89% efficiency, 60% win rate vs. EXPNeuralUCB's 78%, 5%
- **Thompson Sampling:** 88.2% efficiency with **3× speedup** (3.52h vs 11.62h)
- **Fixed Allocator**
Trap: 6.4% generalization penalty under dynamic

allocation

4. Specialized Evaluation Extensions (Papers 4a & 4b)

These intermediate phases validated cross-scenario robustness:

4a. Stochastic_vs_Advers_Eval_Framework

Introduced 5 adversarial regimes (Baseline, Stochastic, Markov, Adaptive, OnlineAdaptive) to quantify variance and vulnerability across conditions.

4b. Hybrid_MABs_Eval_Framework

Focused on hybrid combinations (CPursuit+Thompson), revealing consistent dominance in high-variance environments. Together, these extensions discovered attack-allocator synergies (+2.7% to -26.5%) that inform deployment strategies.

5. Dynamic Routing Synthesis (Paper 5 – Current Framework)

Integrated findings from Papers 1-4 into full-stack routing:

- Scenario-aware strategy pairing (CPursuit+Thompson for OnlineAdaptive)
- Resource-bound variants (1T, 1.5T, 2T capacity scaling)
- Attack-allocator interaction rules for threat-based deployment

How This Differs from Existing Stochastic MAB Work

1. **Multi-scenario stress testing:** Five adversarial models reveal brittleness invisible in single-environment evaluation
2. **Cross-allocator generalization:** EXPNeuralUCB succeeds under Fixed+Markov (original paper's exact conditions) but wins only 1/20 scenario-allocator combinations overall
3. **Allocator-algorithm synergies:** Markov+Dynamic UCB improves +2.7%; Adaptive+Random collapses -26.5%, creating pairing rules
4. **Efficiency under constraint:** Thompson Sampling achieves Fixed-level efficiency while cutting

runtime by 3×—not captured in single-allocator benchmarks

Primary Research Thrusts

Beyond specialized MAB path planning:

- **Evaluation**
methodology: Exposes systematic bias in fixed-allocation testing (standard practice including the quantum paper), quantified via 6.4% generalization penalty metric
- **Co-design**
principles: Allocators and algorithms must be tuned jointly—optimal pairs (CPursuit+Thompson) achieve 97-98% peak efficiency
- **Deployment**
taxonomy: Four winner types (scenario-specific, gap-reduction, efficiency, overall robustness) distinguish production contexts

One-Page Summary

Context-aware Pursuit algorithms with probabilistic allocation achieve superior generalization (89% efficiency, 7.9% CV, 72.9% floor) vs. exponential weighting with fixed allocation (78%, 10.9% CV, 54% floor). Single-allocator evaluations mask critical failure modes—our framework quantifies this bias and provides deployment-ready guidelines for robust quantum routing and general MAB systems under realistic threat conditions.

This version is missing the top papers testbeds integration, as discussed in our previous communication. I am happy to expand on allocator experiments or threat-based recommendations in Overleaf if needed.

Questions

A couple of quick scheduling questions:

- **Hour logging:** I stopped logging hours when the semester ended last week. Should I continue tracking hours, or pause until Spring semester begins?
- **Spring continuation:** As I finalize my Spring schedule, can you confirm whether the GA position continues through Spring semester?

Best,
Piter

On Fri, Dec 26, 2025 at 2:39 PM Daniel Krutz <dxkvse@rit.edu> wrote:
Thanks for the updates and it sounds promising! A few things:

1. Is there an updated Overleaf document that is currently being updated that I should/can look at? I am just wondering as they're a bunch of projects and I want to make sure that I am looking at the correct one.
2. You can include this in the paper, but can you answer a few things:
 1. Do you have a short writeup comparing your work to existing works that use MABs to address stochastic scenarios? What makes your work better/different?
 2. Briefly mention the primary thrusts of your work. For example, do you *just* utilize a special kind of MABs for path planning?

At most, I'd envision the above writeup to be 1 page at most. Just keep it short/to the point.

Thanks
-Dan

On Dec 25, 2025, at 3:26 AM, Piter Garcia (RIT Student) <pzg8794@rit.edu> wrote:

Hi Dan and Devroop,

Merry Christmas!

Quick update on testbeds integration progress following our meeting:

Paper 2 Testing (Default vs Dynamic Allocators)

- Default allocator: Stochastic avg efficiency 92.8%, CPursuit/iCPursuit lead (93.0-93.4% on individual trials)
- Dynamic allocator: Stochastic avg efficiency 92.7%, similar winners (CPursuit leads most trials)
- Markov adversarial results show efficiency drop to ~60% (59-62% range) - highlights allocator impact under attacks
- Key finding: Dynamic allocator provides marginal improvement in Markov scenarios (59-62% vs lower with Fixed)

Paper 7 Testing (Baseline with 100 frames, Dynamic with 50 frames)

- Default (100 frames): EXPNeuralUCB leads across all scenarios (54.7% Stochastic, 41.7% Markov, 42.3% Adaptive)
- Dynamic (50 frames): Mixed results - GNeuralUCB and iCPursuit alternating wins in Stochastic; iCPursuit leads Markov (63-65%)

- In Paper 7, smaller frame sizes indicate K (arms per path) may be limiting exploration - worth investigating
- Performance is lower (40-55% range) vs Paper 2 (92-93%), suggesting frame/capacity dependency

Dynamic Physics Implementation

- Generalized `get_physics_params_paper12()` ready for arbitrary `num_paths/arms_per_path`
- Successfully removes hardcoded '4' constraint - now supports 50+ paths scalably
- Implementation tested in test pipeline; no shape inconsistencies reported

Next Steps & Timeline

- Paper 2 results are solid; moving to running our baseline with this new testbed
- Paper 7 results seem solid; moving to running our baseline with this new testbed
- Paper 12 integration is done, and tests are running
- README contribution guide in progress
- The master datasets notebook is being updated as new tests are run

Excited about the progress we're making. As soon as this is done this week, I will quickly add and test epsilon + NeuralUCB.

Best,
Piter

```
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